Fundamental Physics 1: Mechanics

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Evaluation

- Attendance/Attitude: 10 %
- Exercise: 10 %
- Mid-term test: 30 %
- Final exam: 50 %
- Class exercises/Active students: bonus mark

Contents

1. Kinematics : How to describe motions ?
   - *Motions in one dimension*
     - Displacement, trajectory, velocity and acceleration
     - Motion with constant velocity; Motion with constant acceleration
     - Graphical analysis of motion
     - Free fall motion under gravity and effect of air resistance to the free fall motion
   - *Motions in three dimensions:*
     - Vector and scalar physical quantities
     - Projectile motion
     - Relative motion in different reference frames
2. Dynamics: From force to motion
   - Newton's laws of motion
   - Everyday forces
   - How to apply the laws of motion ?
3. Chapter 3: Circular motion
   - Kinematics of uniform circular motion
   - Centripetal acceleration
4. Chapter 4: Conservation of momentum and mechanical energy
   - Kinetic energy, mechanical work, and power
   - The work-kinetic energy theorem
   - Potential and mechanical energies

Major sources

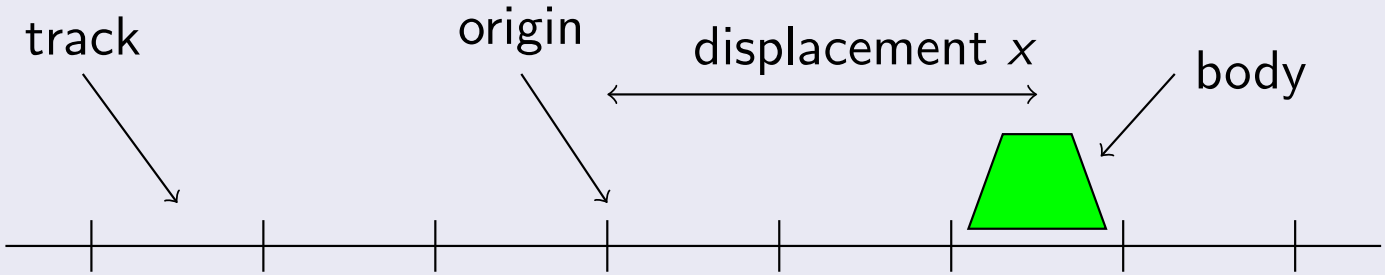
Major sources

1. Halliday and Resnick, Fundamentals of PHYSICS 10th Edition, Jearl Walker
2. G.R. Fowles, Third edition (Holt, Rinehart, & Winston, New York NY, 1977).

Chapter 1: Overview of Classical Mechanics (CM)

- CM is the study of the motion of bodies in accordance with the general principals first enunciated by Issac Newton (1687)
- CM has many applications in different areas of science such as **astronomy**, **Chemistry**, **Geology**, and **Engineering**
- Types of motion studied in CM:
  - **Translational motion**: a body shifts from one point in space to another
  - **Rotational motion**: an extended body changes orientation, with respect to other bodies in space, without changing position
  - **Oscillatory motion**: continually repeats in time with a fixed period
  - **Circular motion**: a body executes a circular orbit about another fixed body.

Displacement



- Distance travelled between t_1 and t_2 (if the direction of motion does not change)

$$S = |x(t_2) - x(t_1)|$$

Velocity

- Velocity is the rate of change of displacement with time
- Average velocity:

$$\bar{v} = \frac{x(t_2) - x(t_1)}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$

- Instantaneous velocity:

$$v = lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

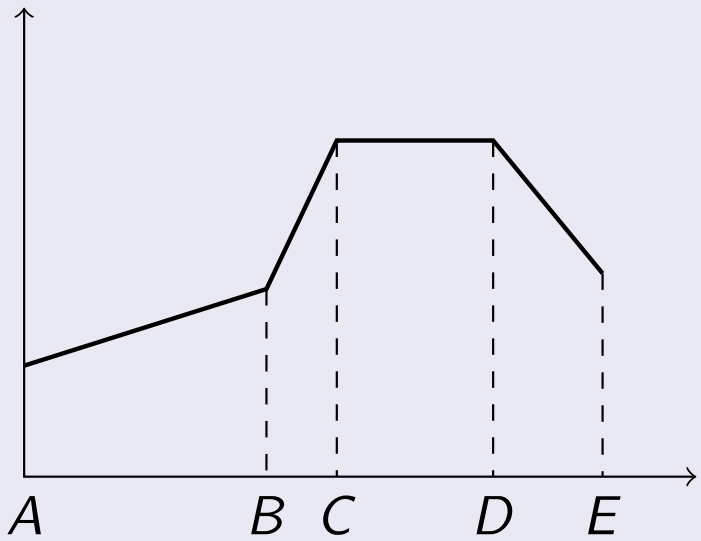
- Average speed: the ratio of the total distance it travels to the total time it takes to travel that distance.
- Instantaneous speed $= |v|$

Chapter 1: Motion in one dimension (3)

Worked example:

1. Given $v = const$, find the dependence of displacement x on time t?
2. Plot the graph of displacement versus time for a body moving with constant velocity?

Motion with constant velocity



- The graph of x versus t is a straight line : $x(t) = x_0 + v.t$
- $v = \Delta x/\Delta t \rightarrow$ gradient of the graph/straight line.
- x increases $\rightarrow v > 0$ OR object moving to the right,
- x decreases $\rightarrow v < 0$ OR object moving to the left,
- $x = \text{const} \rightarrow v = 0$ OR object remains at rest.

Acceleration

- Acceleration is the rate of the change of velocity with time
- Average value:

$$\bar{a} = \frac{\Delta v}{\Delta t}$$

- Instantaneous value:

$$a = lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Acceleration and deceleration

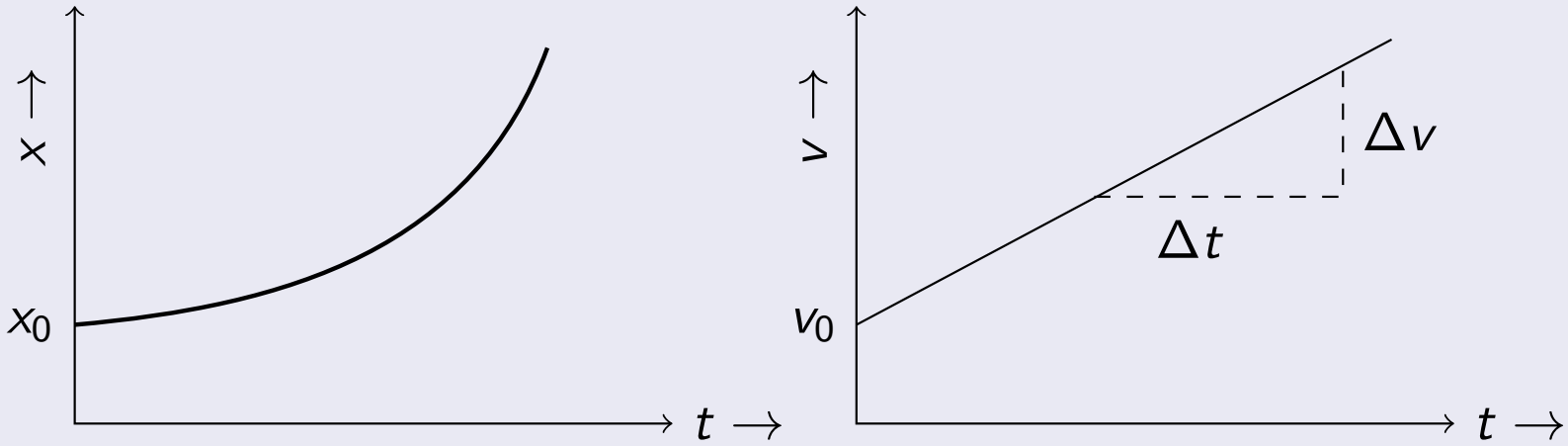
Acceleration and deceleration

- When the speed increases, we have the so-called "acceleration" and acceleration and velocity is in the same direction ($a.v > 0$).
- When the speed decreases, acceleration and velocity is in opposite direction ($a.v < 0$) and we have so-called "deceleration" or "retardation"

Equation of kinematics (for motion with a constant acceleration)

- $v(t) = v_0 + a.t$
- $x(t) = x_0 + v_0.t + a.t^2/2$
- $v(t)^2 - v_0^2 = 2.a.(x(t) - x_0)$

Motion with constant acceleration



- $a = dv/dt \rightarrow dv = a.dt \rightarrow v(t) = v_0 + a.t \rightarrow$ straight line
- $dx = v.dt \rightarrow x = x_0 + v_0.t + a.t^2/2 \rightarrow$ parabola.
- $S = x - x_0 = v_0.t + a.t^2/2 \rightarrow$ the net distance traveled after t seconds
- $v^2(t) = v_0^2 + 2.a.S$

Motion in one dimension: Graphical analysis of motion (1)

Displacement - time graph

- Gradient of graph is the velocity of the body.
- If the gradient of graph is positive $\Rightarrow$ direction of motion is in the positive direction of displacement x.
- If the gradient of graph is negative $\Rightarrow$ direction of motion is in opposite direction of the displacement x.

Displacement - time graph

- Horizontal line – *body at rest*.
- Straight line with positive gradient – *uniform velocity*.
- Straight line with negative gradient – *uniform velocity in the opposite direction*.
- Curve – *non-uniform velocity*

Motion in one dimension: Graphical analysis of motion (2)

Velocity - time graph

- Gradient of graph is the acceleration of the body.
- Area under the graph gives the distance of the body.

Velocity - time graph

Examples: sketch graphs of $(v - t)$ to describe the following motions:

1. Body at rest
2. Body moves with uniform velocity of 5 m/s.
3. Body moves from rest with uniform acceleration.
4. Initial velocity $\neq 0$.
5. Graph with negative gradient.
6. Non-uniform acceleration
   - Acceleration increases
   - Acceleration decreases
   - Deceleration increases
   - Deceleration decreases

Motion in one dimension: Free fall under gravity (1)

- The gravitational force of the Earth pulls all objects towards the centre of the Earth.
- A body is in free fall motion if it moves under the influence of the Earth's gravity alone.
- Without air resistance, a body in free fall has an acceleration known as the acceleration of free fall g.
- Galileo: all bodies in free-fall close to the Earth surface accelerate vertically downward with the same acceleration: $g = 9.81 m/s^2$

Motion in one dimension: Free fall under gravity (2)

???? Sketch **velocity - time graph** and **distance - time graph** for a body under free fall from rest.

Motion in one dimension: Free fall under gravity (3)

??? Sketch **velocity - time graph** and **distance - time graph** for a body projected vertically upwards.

Chapter 1: Effect of air resistance to free-fall motion

- When an object falls through the air, it experiences air resistance, which opposes its motion.
- Because if air resistance, the acceleration of an object falling in air is less than the acceleration of free fall.
- The air resistance increases when the velocity of the object increases.
- For an object which is released from rest and falls in air:
  - Initially its velocity increases,
  - As its velocity increases, the air resistance increases,
  - The resultant acceleration of the object decreases.
  - Finally the acceleration becomes zero and it falls at a constant velocity known as the **terminal velocity**

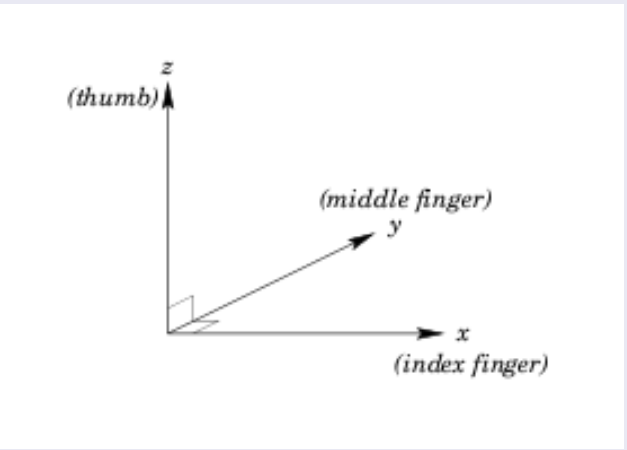
Chapter 1: Motion in three dimensions (1)

To deal with 3-dimensional motion, it is necessary to set up a suitable coordinate system.

Vectors

- Cartesian coordinates

Cartesian coordinate system consists of three mutually perpendicular axes.



- Vector and scalar quantities
  - Scalar quantity is specified by a single value with an appropriate unit and has no direction such as temperature, volume, mass, time
  - A vector quantity has both magnitude and direction. Examples: displacement,velocity, accelerator, force.

Some properties of vector

- Equality of two vectors:
  $\vec{A} = \vec{B} \Leftrightarrow |\vec{A}| = |\vec{B}|$ & $\vec{A}$ and $\vec{B}$ point in the same direction.
- Vector addition
  - Sum of two vectors is the vector down from the tail of the first vector to the tip of the last vector.
  - Parallelogram rule of addition $\rightarrow \vec{A} + \vec{B} = \vec{B} + \vec{A}$
  - Associative law of addition: $\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$
- Negative of a vector: $\vec{A} + (-\vec{A}) = 0 \rightarrow$ have the same magnitude but point in opposite direction.
- Subtracting vector: $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

Components of a vector and unit vectors

- Components of a vector are the projections of that vector along coordinate axes: $\vec{A} = (A_x, A_y, A_z)$.
- ?? Can the components of a vector ever be greater than the magnitude of the vector?
- hint: $A_x = |\vec{A}|.sin\theta.cos\phi$; $A_y = |\vec{A}|.sin\theta \sin\phi$; $A_z = |\vec{A}|.cos\theta$

Components of a vector and unit vectors

- Unit vectors
  - a unit vector is a dimensionless vector having a magnitude of exactly 1. It is used to specify a given direction and have no other physical significance:
    - $\vec{i} \rightarrow x$; $\vec{j} \rightarrow y$; $\vec{k} \rightarrow z$.
    - $|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$
    - $\vec{i}.\vec{j} = \vec{i}.\vec{k} = \vec{j}.\vec{k} = 0$
    - $\vec{A} = A_x.\vec{i} + A_y.\vec{j} + A_z.\vec{k}$.
  - Vector magnitude: $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
- Scalar multiplication:
  - $\vec{S} = \lambda.\vec{r}$, λ is a number/scalar
  - $\Rightarrow \vec{S} = \lambda(x, y, z) = (\lambda x, \lambda y, \lambda z)$: multiplying all the components of the original.
  - Scalar product of two vectors:

$$\vec{A}.\vec{B} = A_x.B_x + A_y.B_y + A_z.B_z$$

Vector displacement, vector velocity and vector acceleration

- Vector displacement: $\vec{r}(t) = [x(t), y(t), z(t)]$
- Vector velocity:

$$\vec{v}(t) = lim_{\Delta t \to 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} = \frac{d\vec{r}}{dt}$$

$$v_x = \frac{dx}{dt}; \quad v_y = \frac{dy}{dt}; \quad v_z = \frac{dz}{dt}$$

- Vector acceleration:

$$\vec{a}(t) = lim_{\Delta t \to 0} \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}$$

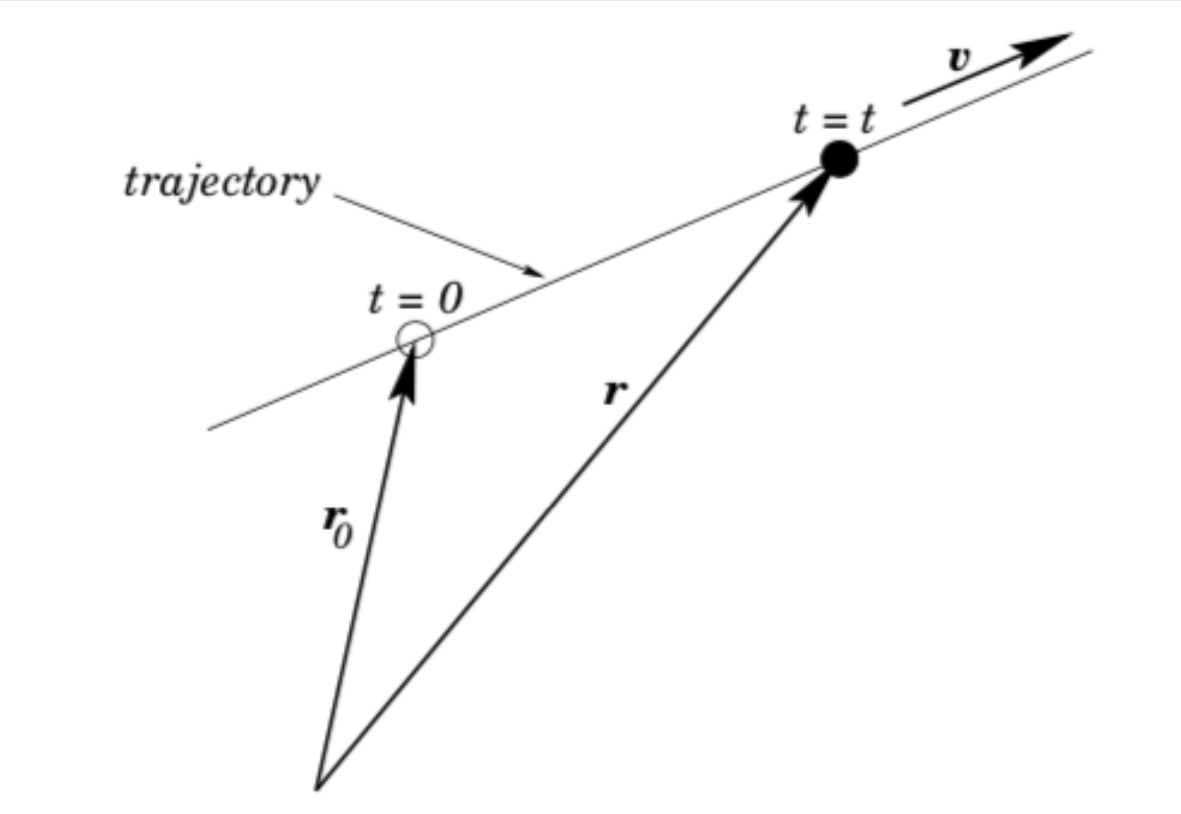
$$a_y = \frac{dv_y}{dt} = \frac{d^2y}{dt^2}$$

$$a_z = \frac{dv_z}{dt} = \frac{d^2z}{dt^2}$$

Motion with constant velocity

- $\vec{r}(t) = \vec{r}_0 + \vec{v}.t$
- The trajectory of the object is parallel to $\vec{v} \Rightarrow$ a straight-line
- In components:

$$x(t) = x_0 + v_x.t$$



Motion with constant acceleration

- $\vec{r}(t) = \vec{r_0} + \vec{v_0}.t + \vec{a}.t^2/2$
- $\vec{S} = \vec{v_0}.t + \vec{a}.t^2/2$
- $\vec{v}(t) = \vec{v_0} + \vec{a}.t$
- $v(t)^2 = v_0^2 + 2\vec{a}.\vec{S}$

In components:

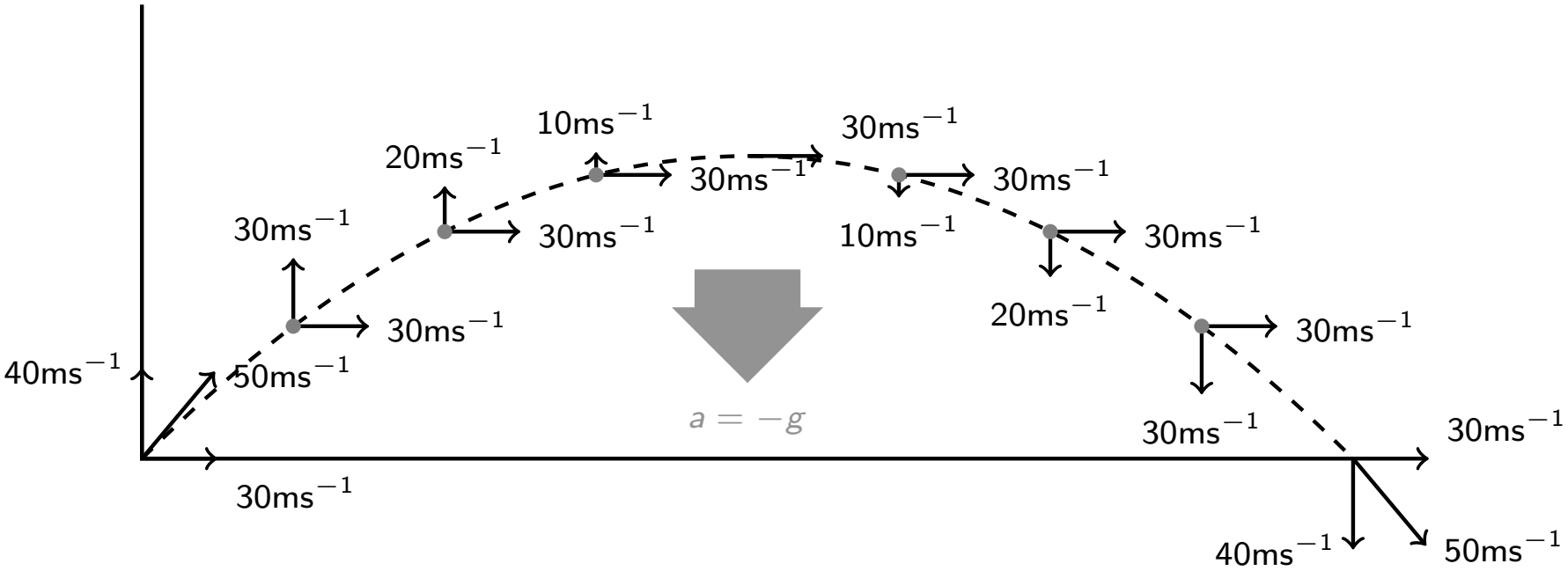
$$x(t) = x_0 + v_{0x}.t + \frac{a_x.t^2}{2}$$

$$v_x(t) = v_{0x} + a_x.t$$

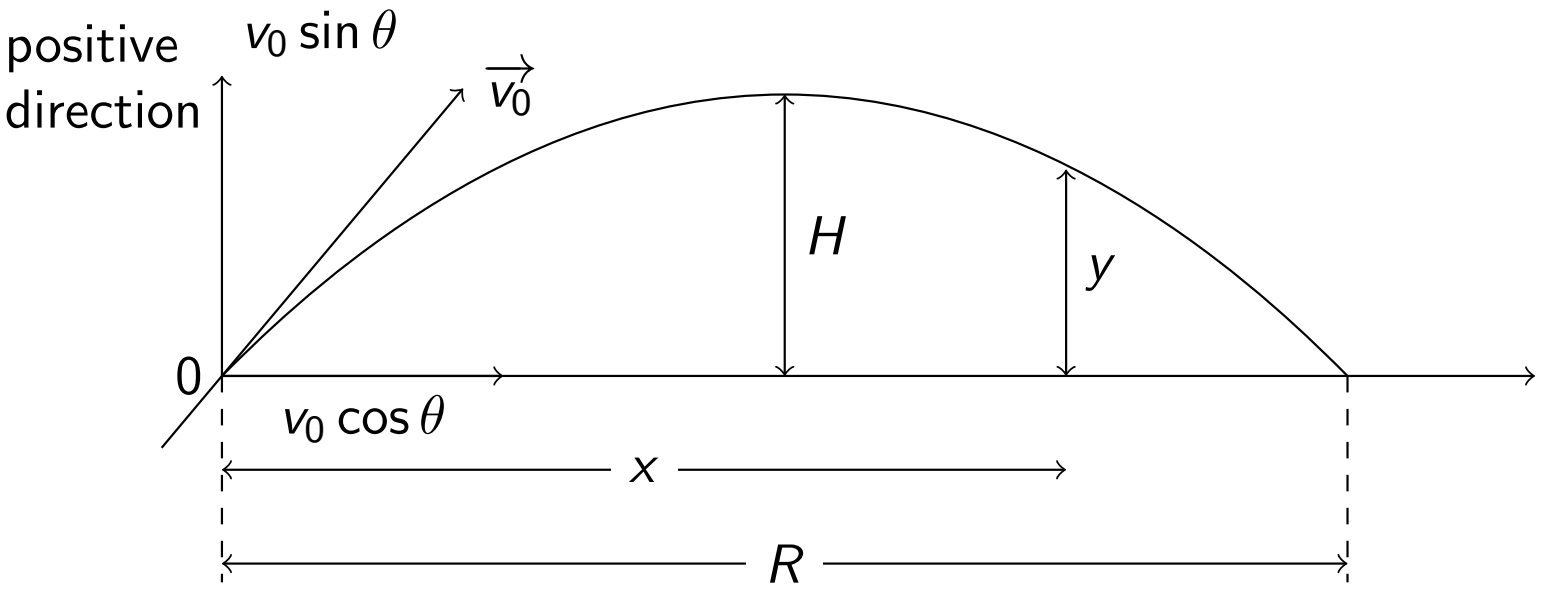
Examples

- Free-fall motion under gravity
- Projectile motion

Chapter 1: Projectile motion (1)



- The horizontal component of the velocity is constant
- The vertical component of the velocity changes
- The vertical component of the velocity is zero at the highest point
- The vertical acceleration $a = -g$ throughout the motion



Projectile motion

- Given: $\vec{a} = (0, -g, 0)$; $\vec{v_0} = (v_0.\cos\theta, v_0.\sin\theta), 0$
- ?? Find horizontal range $R = v_0^2.sin2\theta/g \Rightarrow R_{max} = v_0^2/g$ when $\theta = 45^o$
- ?? Find altitude of the projectile $h = v_0^2.sin^2\theta/2g \Rightarrow h_{max} = v_0^2/2g$ when $\theta = 90^o$

Relative motion

- $\vec{v}_{12} \rightarrow$ velocity of object (1) with respect to frame (2)
- $\vec{v}_{23} \rightarrow$ velocity of frame (2) with respect to frame (3)
- $\vec{v}_{13} \rightarrow$ velocity of object (1) with respect to frame (3)
- Relative velocity: $\vec{v}_{13} = \vec{v}_{12} + \vec{v}_{23}$
- Relative position/displacement: $\vec{r}_{13} = \vec{r}_{12} + \vec{r}_{23}$
- Relative acceleration: $\vec{a}_{13} = \vec{a}_{12} + \vec{a}_{23}$

Chapter 2: Forces and Newton's laws of motion (1)

Forces

- A force is a push or a pull
- Force is a vector quantity. It has both magnitude and direction
- The unit of force is newton (N)
- In general, a body can have several forces acting in it at the same time.

Balanced forces and Newton's first law of motion

- No resultant external forces means that the external forces on the body are balanced.
- If the motion of a given body is not disturbed by external influences then that body moves with constant velocity
- $\vec{r} = \vec{r}_0 + \vec{v}.t, \quad \vec{r}_0 \quad , \quad \vec{v}$ are constant vectors
- if $\vec{v} \neq 0$ (the object is moving), object continues to move in a straight line with constant speed.
- If $\vec{v} = 0 \rightarrow$ the body simply remains at rest.

Unbalanced forces

- Forces acting on a body are unbalanced if the resultant force is not zero.
- Unbalanced forces can change the motion of an object. Unbalanced forces can cause:
  - a stationary object to move
  - the speed of an object to increase
  - the speed of an object to decrease
  - the direction of motion to change.

Newton's second law of motion

- Momentum: $\vec{p} = m.\vec{v}$
- Second law:
$$\frac{d\vec{p}}{dt} = \vec{F} \rightarrow \text{net influence/force}$$
- $m = const \Rightarrow \vec{F} = m.\vec{a}$

Newton's second law of motion

The rate of change of momentum of an object is directly proportional to the applied force and it occurs along the direction of the force.

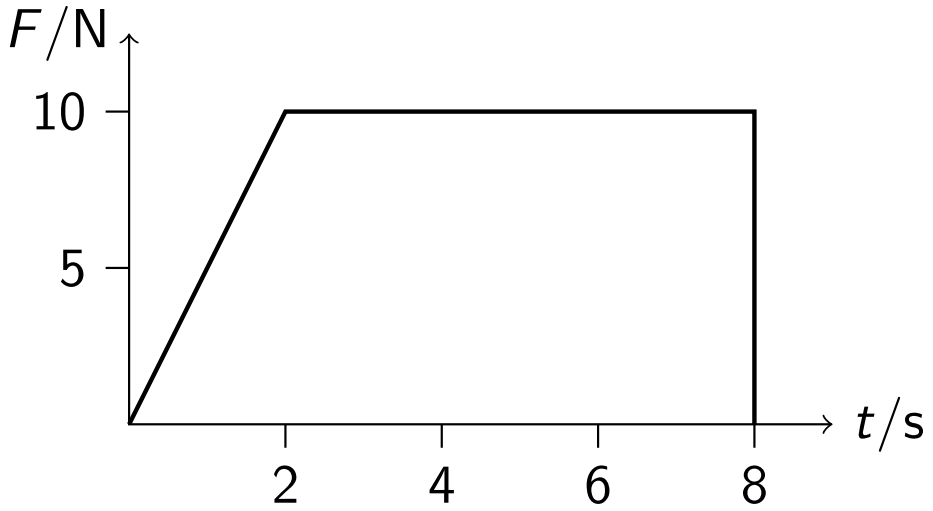
$$F = \frac{d}{dt}(mv)$$

$$\int_{t_1}^{t_2} F.dt = mv - mu$$

- $\int_{t_1}^{t_2} F.dt \Rightarrow$ *impulse* of the force $F \rightarrow$ the change in momentum.
- The impulse of a constant force is the product of the force and the time duration that it acts.
- *Impulse* of the force $F =$ area below $(F - t)$ graph between t_1 and t_2

Chapter 2: Worked example

A body of mass 3 kg experiences a force F which varies with time t as shown in the Figure below. What is the increase in momentum of the object after 8 s ?

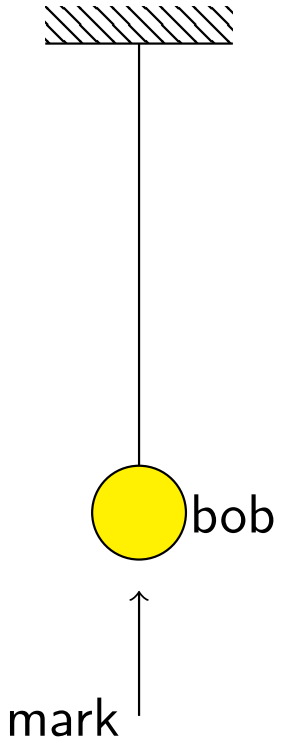


A plumb-line is hung from the roof of a train carriage. The bob is vertically above a mark when the train is stationary. Draw figures to show the position of the bob relative to the mark when

(a) the train accelerates uniformly,

(b) the train moves with uniform velocity,

(c) the train retards uniformly.





Newton's third law of motion

- $\vec{f}_{ab}$: force exerted on (a) by (b)
- $\vec{f}_{ab}$: force exerted on (b) by (a)
- Newton's third law: $\vec{f}_{ab} = -\vec{f}_{ba}$
- Why Newton's third law? $\rightarrow$ No self generated forces.

- **Gravitational force**: $\vec{f_g} = m.\vec{g}$
- **The normal force/normal reaction and weight**:
  - Weight: the weight W of a body is the magnitude of the downward force it exerts on any object which supports it.
  - Normal force: under the influence of the weight of the object, obeying the Newton's third law, the surface of the supporting object will act back on the surface of the object $\Rightarrow$ this called **"normal reaction"** or **"normal force"**
  - $\Rightarrow$ **"Normal force"** is a force exerted by one surface on another in a direction perpendicular to the surface of contact.
- The magnitude of the **weight** = the **normal force**.
- Relation between **mass** and **weight**: $W = m.g$ in case of accelerator of the frame = 0 and $\vec{f_g} = m\vec{g}$ is perpendicular to the surface.

Everyday forces: Weight

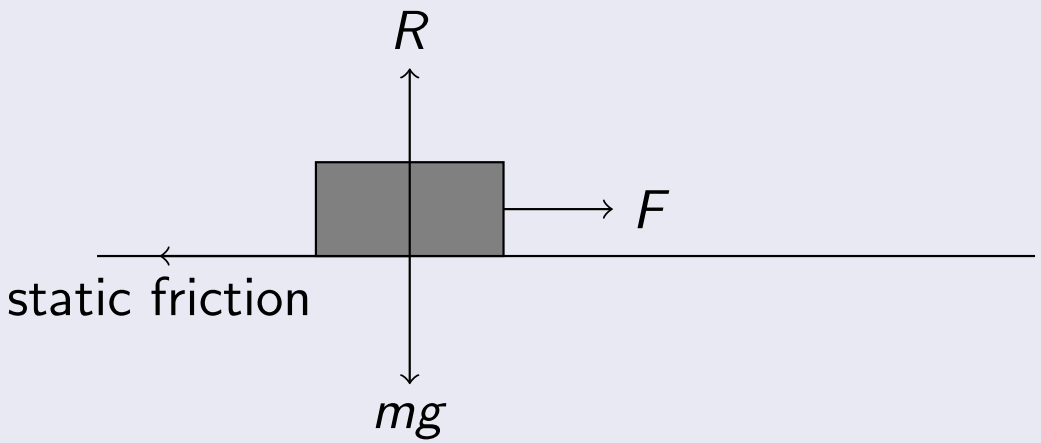
- Find weight of an object resting on a table?

Everyday forces: Weight

- Find weight of an object of mass m resting on the floor of an elevator moving with acceleration $\vec{a}$?

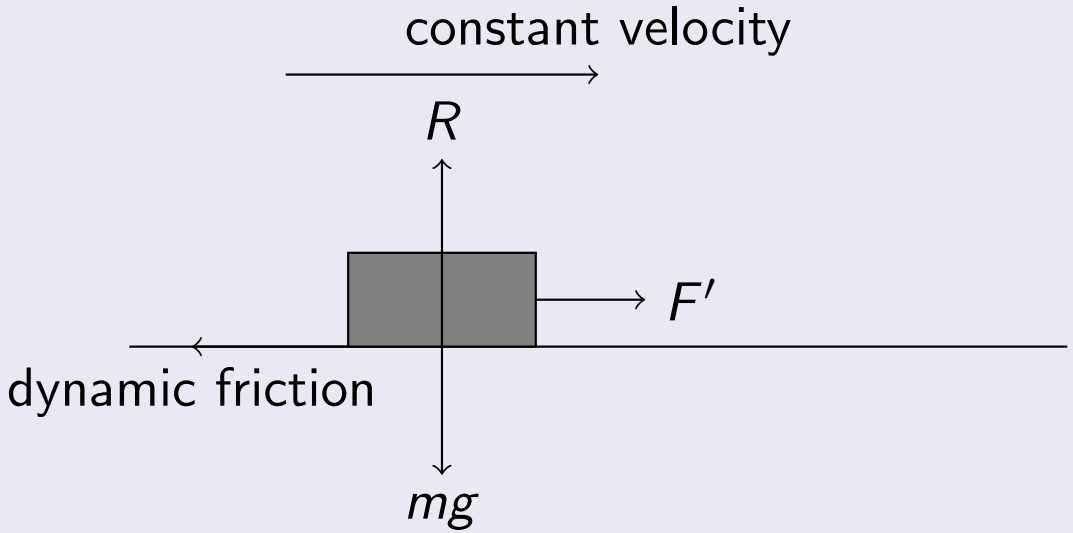
Static friction

- The magnitude of the static friction is equal to applied force but opposite direction to the applied force.
- The maximum static friction occurs just before motion takes place. It is known as *limiting static friction*



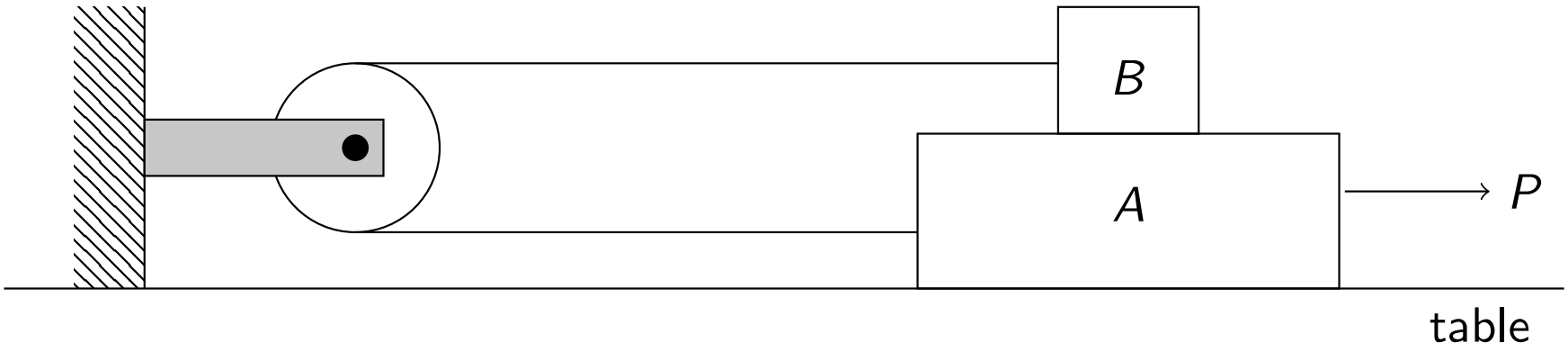
Kinetic/dynamic friction

$f = \mu.R_n$, where μ : coefficient of friction and R_n is normal reaction of the surface.



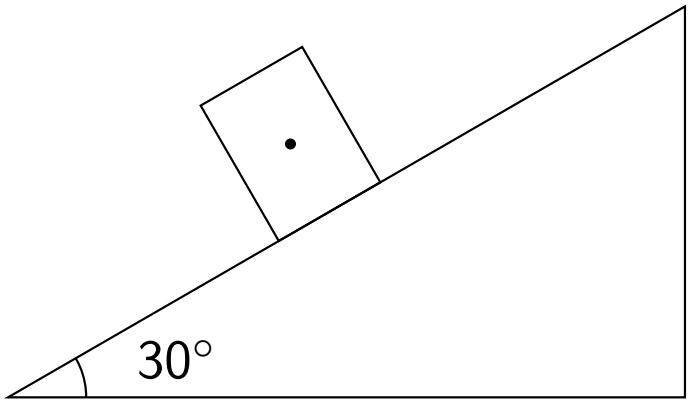
Chapter 2: Worked example

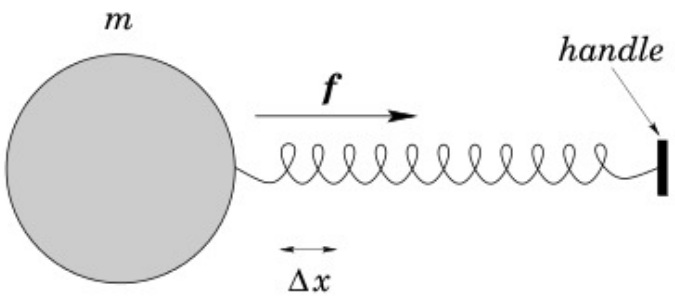
The below figure shows a wooden block B at rest on another block A. Both blocks are connected by a string which goes round a fixed smooth pulley. The dynamic friction between block A and the table is 14 N and that between block B and A is 10 N. What is the horizontal force P which is required to move block A with uniform velocity?



Chapter 2: worked example

A metal block of mass 4 kg is on a plane inclined at an angle of 30° to the horizontal. The block slides down the plane from rest and travels a distance of 1.0 m in 2.0 s. What is the frictional force between the block and the inclined plane when the block is slipping?





- The force exerted on an object by a coiled spring:

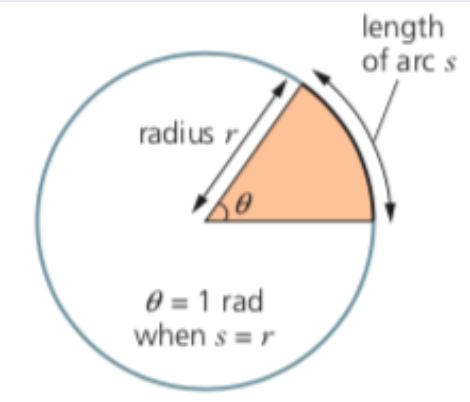
$$\vec{f} = -k.\Delta\vec{l}; \qquad (\Delta\vec{l} \equiv \Delta\vec{x})$$

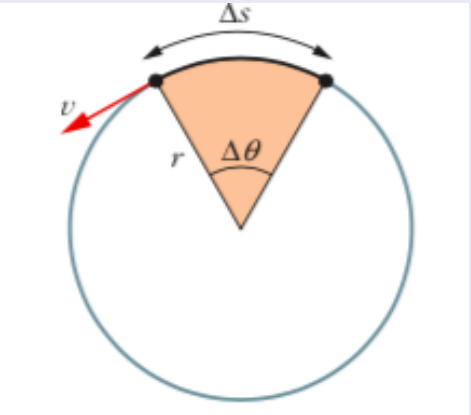
  - $\Delta\vec{l} = \vec{l} - \vec{l_0} \rightarrow$ extension of the spring
  - $\vec{l} \rightarrow$ actual length; $\vec{l_0} \rightarrow$ natural length.
- Application of Hooke's law: to quantify the magnitude and direction of the force exerted on a given body by means of a spring
  - $F_0 = 1N = 1kg.m/s^2 = k.\Delta l_0$, $\Delta l_0 \rightarrow$ critical extension
  - $F \propto \Delta l = \alpha.\Delta l_0 \Rightarrow \alpha$ is the value of the force we want to quantify.
  - Strain: $s = \Delta l / l_0$

Kinematics of uniform circular motion

- Radian measure and angular displacement:
  One radian (rad) is defined as the angle subtended at the centre of a circle by an arc equal in length to the radius of the circle.

$$\theta = \frac{\text{length of arc}}{\text{radius of circle}} = \frac{s}{r}$$





- Angular speed: $\omega = \Delta\theta/\Delta t$
- Linear speed: $v = \omega.r$
- Period: $T = 2\pi/\omega$
- Frequency: $f = 1/T$

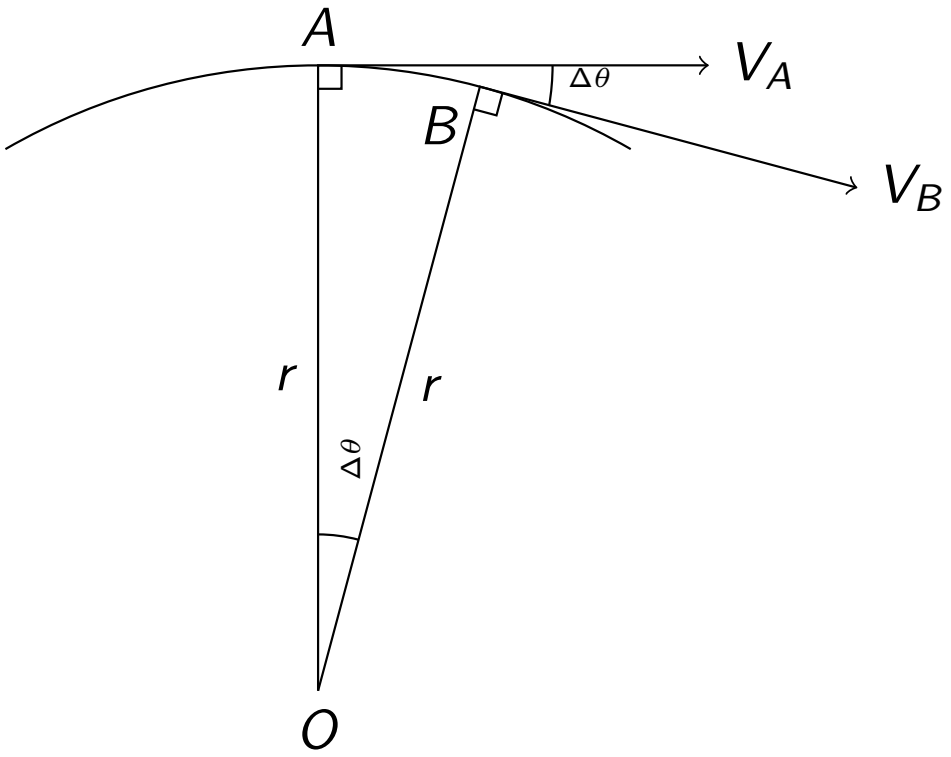
Worked example

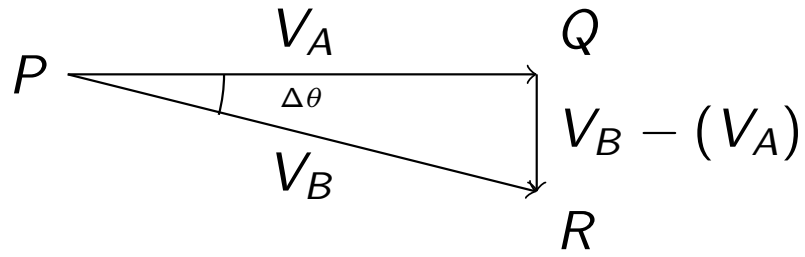
A phonograph record rotates at $33\frac{1}{3}$ revolutions per minute.

(a) Express the angular velocity of the record in radiant per second.

(b) What is the speed of a lady bird sitting on the record at a distance of

- 5.0 cm
- 10.0 cm from the center?





- Centripetal acceleration: toward the centre of the circle.
- Magnitude: $a = \Delta v/\Delta t = \nu(\Delta\theta)/\Delta t = \nu\omega$
- Centripetal force:

$$F = m\nu\omega = mr\omega^2 = \frac{m\nu^2}{r}$$

Worked example

The drum of a spin dryer has a radius of 50cm and rotates at 720 revolutions per minute.

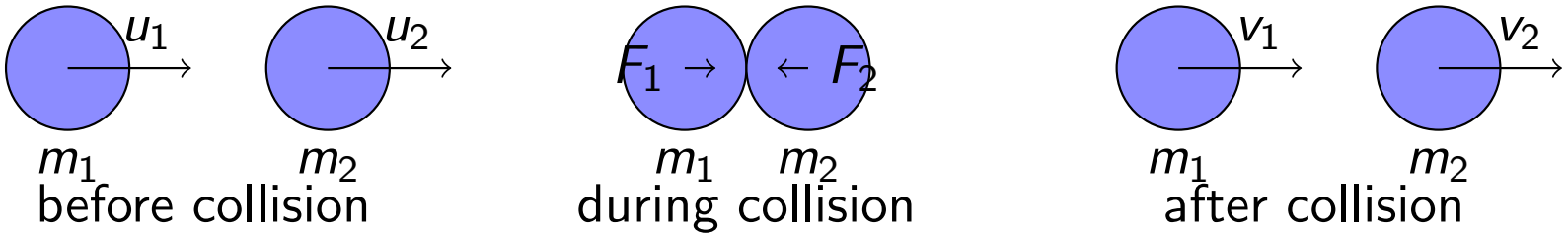
1. Show that the angular speed of the drum is about 75 rad s^{-1}
2. Calculate, for a point on the edge of the drum:
   - its linear speed
   - its acceleration towards the centre of the drum.

Worked example

A rope is tied to a bucket filled with water, and the bucket is swung in a vertical circle of radius 1.2 m. What must be the minimum speed of the bucket at the highest point of the circle if the water is to stay in the bucket throughout the motion?

A collision is a phenomenon which has these features:

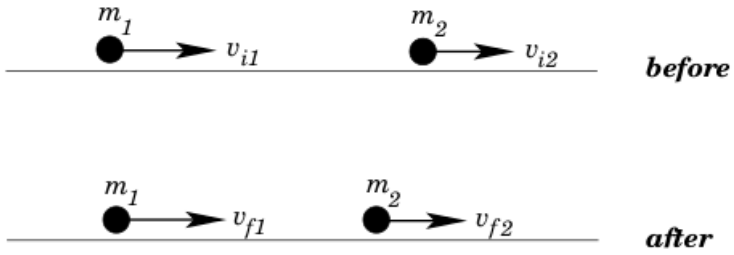
1. it occurs in a short time interval,
2. what happens after the collision differs from what happens before the collision,
3. the colliding bodies may be assumed to constitute a closed system,
4. momentum and energy are conserved during the collision.



Vector sum of linear momentum after collision = vector sum of linear momentum before collision

$\Rightarrow$ **Linear momentum is conserved!**

- m_1 and m_2 collide in 1-dimension
- ignore zero net external force in the direction of moving
  - Momentum conservation gives



$$m_1\vec{v}_{i1} + m_2\vec{v}_{i2} = m_1\vec{v}_{1f} + m_2\vec{v}_{f2} \quad (1)$$

  - In 1-dimention: $m_1 v_{i1} + m_2 v_{i2} = m_1 v_{f1} + m_2 v_{f2}$

Types of collision

- Elastic collision: Total kinetic energy of the two colliding objects is conserved

$$\frac{1}{2}m_1 v_{i1}^2 + \frac{1}{2}m_2 v_{i2}^2 = \frac{1}{2}m_1 v_{f1}^2 + \frac{1}{2}m_2 v_{f2}^2$$

- Inelastic collision: some fraction of the initial kinetic energy of the colliding objects is usually converted into some other form of energy.
- Totally inelastic collision: two objects stick together after the collision $\Rightarrow v_{f1} = v_{f2}$

There are many types of collision

- Elastic collision: Total kinetic energy of the two colliding objects is conserved
$$\frac{1}{2}m_1 v_{i1}^2 + \frac{1}{2}m_2 v_{i2}^2 = \frac{1}{2}m_1 v_{f1}^2 + \frac{1}{2}m_2 v_{f2}^2$$
- Inelastic collision: some fraction of the initial kinetic energy of the colliding objects is usually converted into some other form of energy.
- Totally inelastic collision: two objects stick together after the collision $\Rightarrow v_{f1} = v_{f2}$

Newton's law of restitution

- **When two bodies collide, relative velocity after collision =** $-e\times$ **(relative velocity before collision)**.
- e ; $(0 \leq e \leq 1)$ is coefficient of restitution
- **Elastic collision:** $e = 1$
- **In-elastic collision,** $0 < e < 1$
- **Total in-elastic collision,** $e = 0$

- Momentum conservation

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$

- Using Newton's law of restitution

$$v_1 = v_2 = -e(u_1 - u_2)$$

- Results:

$$v_1 = \frac{2m_2 u_2}{m_1 + m_2} + \frac{(m_1 - m_2)u_1}{m_1 + m_2}$$

$$v_2 = \frac{2m_1 u_1}{m_1 + m_2} + \frac{(m_2 - m_1)u_2}{m_1 + m_2}$$

$$v_1 = \frac{2m_2u_2}{m_1+m_2} + \frac{(m_1-m_2)u_1}{m_1+m_2}$$

$$v_2 = \frac{2m_1u_1}{m_1+m_2} + \frac{(m_2-m_1)u_2}{m_1+m_2}$$

We consider 3 following cases:

1. $m_1 = m_2$ In an elastic collision between two bodies of the same mass, the velocities of the bodies interchange after collision
2. $m_2 >> m_1$ and m_2 at rest
   $v_1 = -u_1$ and $v_2 = 0$
   The small mass m_1 rebounces the same speed, whereas the large mass m_2 remains at rest.
3. m_1 is very large compared to m_2 ($m_1 >> m_2$) and m_2 initially at rest.

$$v_1 = u_1, \quad \text{and} \quad v_2 = 2u_1$$

- We have: $v_f = v_{f1} = v_{f2}$
- $\Rightarrow m_1 v_{i1} + m_2 v_{i2} = m_1 v_{f1} + m_2 v_{f2} = (m_1 + m_2) v_f$

$$\Rightarrow v_f = \frac{m_1 v_{i1} + m_2 v_{i2}}{m_1 + m_2} = v_{cm}$$

two objects remain stationary in the center of mass velocity frame.

- Suppose that $v_{i2} = 0$

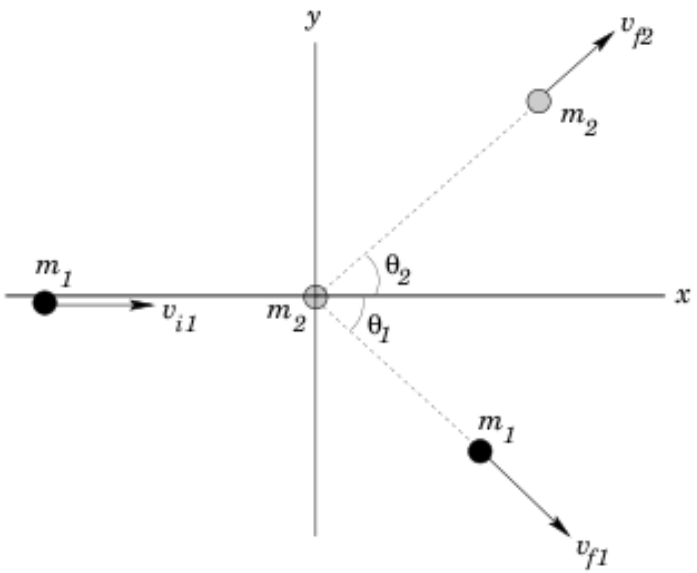
$$v_f = \frac{m_1}{m_1 + m_2} v_{i1}$$

$\rightarrow$ The fractional loss in kinetic energy of the system due to the collision is

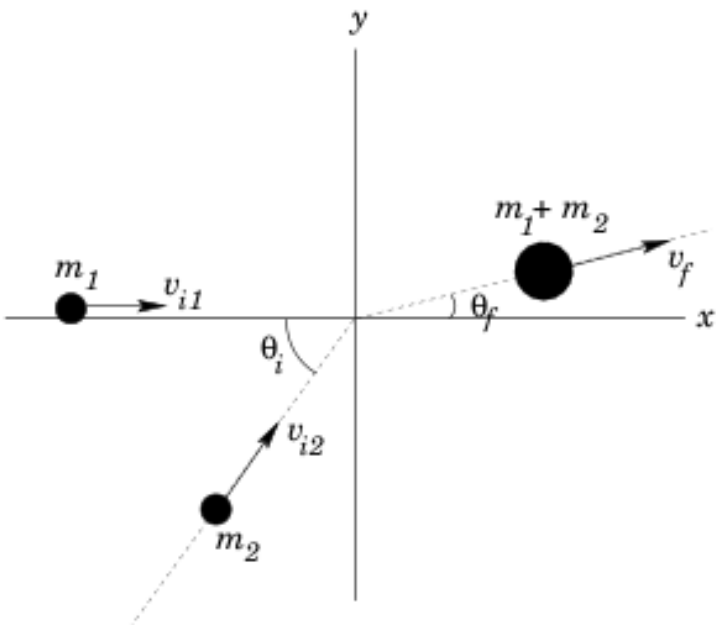
$$f = \frac{K_i - K_f}{K_i} = \frac{m_1 v_{i1}^2 - (m_1 + m_2) v_f^2}{m_1 v_{i1}^2} = \frac{m_2}{m_1 + m_2}$$

  - If $m_2 \ll m_1 \Rightarrow$ the loss in energy is small
  - If $m_2 \gg m_1 \Rightarrow$ the loss is almost 100 %

- Suppose $v_{i2} = 0$; $m_1, m_2, v_{i1} \rightarrow$ known
- Collision is NOT head-on
- Unknown: $v_{f1}, v_{f2}, \theta_1, \theta_2$
- $\Rightarrow m_1 \vec{v}_{i1} = m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2}$ (1)
- (1)/ox:
  $m_1 v_{i1} = m_1 v_{f1} \cos\theta_1 + m_2 v_{f2} \cos\theta_2$
- (1)/oy: $m_1 v_{f1} \sin\theta_1 = m_2 v_{f2} \sin\theta_2$
- **Elastic collision**:
  $m_1 v_{i1}^2/2 = m_1 v_{f1}^2/2 + m_2 v_{f2}^2/2 \Rightarrow$ need one more condition!



- **Totally inelastic colliding**:
  - Unknown: θ_f, v_f
  - $m_1\vec{v}_{i1} + m_2\vec{v}_{i2} = (m_1 + m_2)\vec{v}_f \quad (2)$
  - (2)/ox: $m_1 v_{i1} + m_2 v_{i2}\cos\theta_i = (m_1 + m_2)v_f \cos\theta_f$
  - (2)/oy: $m_2 v_{i2}\sin\theta_i = (m_1 + m_2)v_f \sin\theta_f$

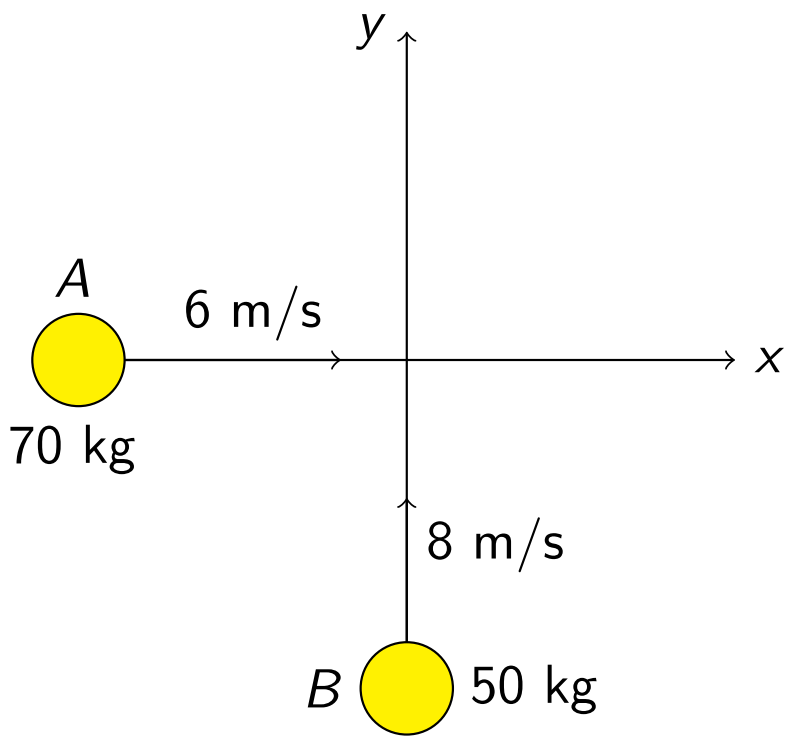


Worked example

Two people A and B of masses 70 kg and 50 kg respectively ski with velocities 6 m/s and 8 m/s along the directions as shown in the figure. They collides and hug each other after collision. What is:

(a) their velocity after collision?

(b) the fraction loss of kinetic energy?



Energy

- Energy is the capacity to do work and is measured in joules (J).
- Different forms of energy: kinetic energy, potential energy, light energy, thermal energy, electrical energy and nuclear energy.
- Kinetic energy is the energy that a body possesses due to its motion

$$K = \frac{1}{2}mv^2$$

- Potential energy is energy stored in an object due to its position and state of shape. Example: chemical, gravitational and elastic potential energy.

Energy conversion

- Energy can be converted from one form to another.
- The efficiency of a device used to convert one form of energy into another:

$$\text{Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$$

An example of energy conservation: energy conservation during free-fall

- $v_2^2 - v_1^2 = 2\vec{a}.\vec{S};\ \vec{a} = \vec{g};\ \vec{S} = \vec{h}_2 - \vec{h}_1$
- $\Rightarrow v_2^2 - v_1^2 - -2\vec{g}.\vec{S}$

$$\Rightarrow \frac{mv_2^2}{2} - \frac{mv_1^2}{2} = -mgh_2 + mgh_1$$

$$\Rightarrow \frac{1}{2}mv_2^2 + mgh_2 = \frac{1}{2}mv_1^2 + mgh_1$$

- **Total energy** E of the mass m is conserved: $E = mv^2/2 + mgh$
  - $K = mv^2/2 \rightarrow$ **kinetic energy**
  - $U = mgh \rightarrow$ **potential energy**
  - E conserved $\Rightarrow \Delta K = -\Delta U$

Work

- $W = \vec{f}.\vec{S} = |\vec{f}|.|\vec{S}|.cos\alpha$: assuming that the force doesn't vary with position.
- **Work-Energy theorem**: ?? prove:

$$\Delta K = W$$

using $v^2 = v_0^2 + 2.\vec{a}.\vec{S}$

- Find back the energy conservation during free-fall using work-energy theorem?

- Work of the force varies with position $\vec{f} = \vec{f}(\vec{r})$
  - $W = \sum_{i=1}^{N} \vec{f}_i.\Delta\vec{r}_i = lim_{N\to\infty} \sum_{i=1}^{N} \vec{f}_i.\Delta\vec{r}_i = \int_A^B \vec{f}(\vec{r}).d\vec{r} \rightarrow$ line-integral
  - Work performed by a 1-dimensional force: From $W = \int_{x_A}^{x_B} f(x)dx$, prove: $W = \Delta K$?

$$f = m\frac{d^2x}{dt^2} \Rightarrow W = \int_{x_A}^{x_B} m\frac{d^2x}{dt^2}.dx = \int_{t_A}^{t_B} m\frac{d^2x}{dt^2}.\frac{dx}{dt}.dt = \int_{t_A}^{t_B} \frac{d}{dt}\left[\frac{m}{2}\left(\frac{dx}{dt}\right)^2\right]$$

$$x(t_A) = x_A,\ x(t_B) = x_B,\ v_A = \left.\frac{dx}{dt}\right|_{t=t_A},\ v_B = \left.\frac{dx}{dt}\right|_{t=t_B}$$

$$\Rightarrow W = \frac{1}{2}m.v_B^2 - \frac{1}{2}mv_A^2 = \Delta K$$

  - The object is acted by many forces: $W_i = \int_A^B \vec{f}_i(\vec{r}).d\vec{r}$

$$W = \sum_i W_i = \sum_i \int_A^B \vec{f}_i(\vec{r}).d\vec{r} = \int_A^B \left(\sum \vec{f}_i(\vec{r})\right).d\vec{r} = \int_A^B \vec{f}(\vec{r}).d\vec{r};$$

$$\vec{f}(\vec{r}) = \sum_i \vec{f}_i(\vec{r})$$

- **potential energy** is the energy associated with a system of objects (consists of two or more objects that exert forces on one another).

Gravitational Potential Energy

- **Gravitational potential energy** is the potential energy of the object - Earth system.
- **gravitational potential energy** is defined as the product of the magnitude of the gravitational force mg acting on an object and the height y of the object

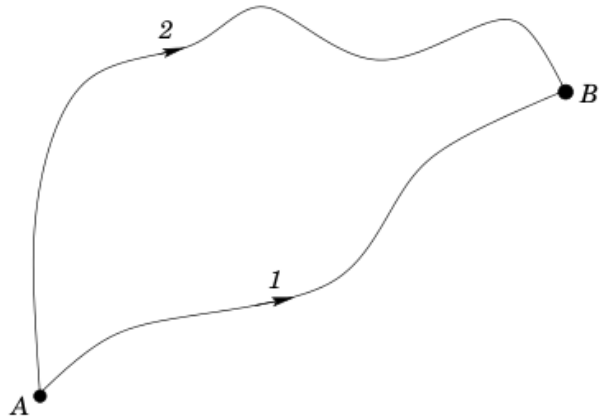
$$U_g = mgh$$

- The work done on any object by the gravitational force is equal to the negative of the change in the system's gravitational potential energy:

$$W_g = U_i - U_f = -\Delta U_g$$

- only the difference in the gravitational potential energy at the initial and final locations that matters
- Horizontal motion does not affect the value of W_g

A non-uniform force-field $\vec{f}(\vec{r})$ acts upon an object which moves along curved trajectory (from point A to point B) labeled by "path 1" and "path 2".



- Work W_1 performed by $\vec{f}(\vec{r})$ along path 1: $W_1 = \int_{A\to B}^{path1} \vec{f}.d\vec{r}$
- Work W_1 performed by $\vec{f}(\vec{r})$ along path 2: $W_2 = \int_{A\to B}^{path2} \vec{f}.d\vec{r}$
- If $W_1 = W_2 \Rightarrow \vec{f}(\vec{r})$ is a **conservative force-field**: the line-integral doesn't depend on the path taken between the end points; Ex: gravitational force.
- If $W_1 \neq W_2 \Rightarrow \vec{f}(\vec{r})$ is a **non-conservative force-field**: the line-integral depends both on the end points A, B and the path taken between them; Ex: frictional force.

Consequences of conservative and non-conservative force-field

- Work done around a close loop for:
  - Conservative force-field:

$$\Delta W = \int_{A \to B} \vec{f}.d\vec{r} + \int_{B \to A} \vec{f}.d\vec{r} = W_1 - W_2 = 0$$

  - Non-conservative force-field:

$$\Delta W = W_1 - W_2 \neq 0$$

- Consider a body moving in a conservative force-field $\vec{f}(\vec{r})$. Pick arbitrarily some point O in this field.
- Define:

$$U(R) = -\int_O^R \vec{f}.d\vec{r} \tag{3.1}$$

  - (3.1)$\Rightarrow U(O) = 0$
  - $U(R)$ defined by (3.1) is unique in conservative force-field since in this field, the work of the field depends only on the fix points but NOT the path taken between them.
  - (3.1) makes no sense in non-conservative force field since there will be an infinite number of different values corresponding to the infinite number of different paths the body coild take between O and R.
- Consider the object move between A and B
  - From the work-energy theorem:

$$\Delta K = \int_A^B \vec{f}.d\vec{r} = \int_A^B \vec{f}.d\vec{r} = \int_O^B \vec{f}.d\vec{r} - \int_O^A \vec{f}.d\vec{r} = -\Delta U$$

$$\Rightarrow E = K(A) + U(A) = K(B) + U(B) = K + U = constant$$

$\Rightarrow$ U **represents some form of "potential energy"**

Chapter 4: Some conclusions about potential energy

- Possible to associate a potential energy with any conservative force-field.
- Any force-field for which we can define a potential energy must necessary be conservative.
- The concept of potential energy is meaning-less in a non-conservative force-field.
- Potential energy define to within an arbitrary additive constant so we can choose an arbitrary point at which $U = 0$.
- Potential energy is not a property of the body BUT a property of the force-field within which the body moves.
- The potential energy function U can be defined such that the work done by a conservative force equals the decrease in the potential energy of the system:

$$W = \int_i^f \vec{F}.d\vec{S} = U_i - U_f$$

- Consider a system consisting of a block plus a spring
- Force: $f = -kx$
- Work: $W = \int_{x_A}^{x_B} f(x)dx = -k\int_{x_A}^{x_B} x.dx = -[kx_B^2/2 - kx_A^2/2]$
- Potential energy: $U(x_B) - U(x_A) = -\int_{x_A}^{x_B} f(x).dx = kx_B^2/2 - kx_A^2/2$

  $\Rightarrow U(x) = kx^2/2$

  $$\Rightarrow f = -\frac{dU}{dx}$$
- In three-dimensional;

$$\vec{f} = -\left(\frac{\partial U}{\partial x}, \frac{\partial U}{\partial y}, \frac{\partial U}{\partial z}\right)$$

- Example in gravitational force-field: $U = mgz$
  $\Rightarrow \vec{f} = (0, 0, -mg)$
- Total energy of the mass: $E = K + U$

- The total mechanical energy of a system remains constant in any isolated system of objects that interact only through conservative forces.

$$K_i + U_i = K_f + U_f$$

- If there are more than one conservative force acting on an object within a system (sum over all conservative forces present):

$$K_i + \sum U_i = K_f + \sum U_f$$

Chapter 4: Work done by non-conservative force

- If there are some of the forces acting on objects within the system are **not conservative**, the total mechanical energy of the system does not remain constant. The change in mechanical energy is equal the total work done by non-conservative forces present.

Work done by an applied force

- Work-kinetic energy theorem: $W_{app} + W_g = \Delta K$
- Gravitational force is conservative $\rightarrow W_g = -\Delta U$
- $\Rightarrow \quad W_{app} = \Delta K + \Delta U$
  $\Rightarrow$ an applied force can transfer energy into or out of the system.

Work done by kinetic friction

- Kinetic friction makes total mechanical energy of the system dissipated (as heat, e.g):

$$\Delta E = \Delta K + \Delta U = \vec{F}_f.\vec{S} < 0$$